

Review Article

Mechanisms of lysosomal tubulation and sorting driven by LRRK2

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Lysosomes are dynamic cellular structures that adaptively remodel their membrane in response to stimuli, including membrane damage. Lysosomal dysfunction plays a central role in the pathobiology of Parkinson's disease (PD). Gain-of-function mutations in Leucine-rich repeat kinase 2 (LRRK2) cause familial PD and genetic variations in its locus increase the risk of developing the sporadic form of the disease. We previously uncovered a process we term LYTL (LYsosomal Tubulation/sorting driven by LRRK2), wherein membrane-damaged lysosomes generate tubules sorted into mobile vesicles. Subsequently, these vesicles interact with healthy lysosomes. LYTL is orchestrated by LRRK2 kinase activity, via the recruitment and phosphorylation of a subset of RAB GTPases. Here, we summarize the current understanding of LYTL and its regulation, as well as the unknown aspects of this process.

Introduction

Lysosomes are catabolic organelles that clear intracellular macromolecules through autophagy, or extracellular cargo via endocytosis. Discovered as acidic compartments full of hydrolases [1], lysosomes were initially thought to exclusively act as degradative stations. Acidification of the lysosomal lumen is critical for lysosomal degradation, as an acidic milieu is required for the proper functioning of many acid hydrolases. Lysosomal acidification is dependent on the large multisubunit mechanoenzyme vacuolar (H^+) ATPase (vATPase) that pumps protons into the lysosomal lumen by consuming ATP [2]. However, recent progress has expanded our view on how lysosomes regulate cellular homeostasis [3]. For example, it has been shown that, (i) lysosomes locally regulate the nutrient-sensing response through the mechanistic target of rapamycin (mTOR) [4]; (ii) lysosomes are well-known calcium (Ca^{2+}) storage sites and therefore regulate Ca^{2+} homeostasis [5]; (iii) lysosomes also control the morphology of other organelles, including mitochondria [6] and the endoplasmic reticulum (ER) [7]; and (iv) lysosomes can bind and transport other cellular compartments to distal regions of the cell [8,9], which is important for mitochondrial homeostasis and axonal maintenance [10]. Overall, lysosomes are crucial organelles for maintaining cellular homeostasis and to preserve cellular health in stressful situations [3,11,12]. Therefore, it is not surprising that lysosomal dysfunction is a hallmark in numerous human diseases, including Parkinson's disease (PD) [13–19].

PD is a neurodegenerative disorder characterized in part by the loss of dopaminergic neurons in the substantia nigra pars compacta and protein deposition pathology in multiple brain regions [20], resulting in a myriad of motor [21] and non-motor symptoms [22]. Nearly 10 million people are currently living with PD worldwide, and this number is expected to increase over the coming decades [23] (parkinson.org). Unfortunately, while there are effective treatments for some symptoms early in the disease course, there are currently no disease-modifying therapies for PD. In terms of mechanisms of disease risk, several genetic causes for PD have been identified in recent years [24]. Among these, coding variants in *Leucine-rich repeat kinase 2* (*LRRK2*) cause familial PD [25,26], and non-coding variation in the gene locus alters the risk of developing the sporadic form of the disease [27,28].

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LRRK2 encodes for a large protein with two enzymatic activities: the ROC–COR bi-domain which can hydrolyse GTP (although with modest kinetics), and the Kinase domain which phosphorylates serine and threonine residues [29]. PD-causing mutations are clustered within these domains, resulting in a toxic protein with hyperactive kinase activity [30]. In terms of cellular functions, several reports have associated LRRK2 with lysosomal biology. Mice lacking LRRK2 develop age-dependent kidney degeneration [31,32], correlated with lysosomal abnormalities in kidney epithelial cells [33]. Indeed, unbiased proteomics have shown dysregulated expression of lysosomal enzymes as an early event in murine LRRK2 knockout kidneys, preceding degeneration [34]. Additionally, pathogenic LRRK2 mutations alter lysosomal morphology and function in certain cell types, including astrocytes [35], fibroblasts [36] and microglia [37]. Taken together, and given the central role of lysosomes in PD pathogenesis driven by multiple genes [16,17,38], it is plausible that the role of LRRK2 at lysosomes may be critical for PD pathogenesis.

We and others have shown that LRRK2 is recruited to the membrane of damaged lysosomes [39–41]. The presence of LRRK2 on lysosomes is exacerbated by lysosomotropic reagents that can selectively accumulate inside lysosomes, thereby changing lysosomal function [42]. Many lysosomotropic reagents induce lysosomal damage and break the lysosomal membrane in a process known as lysosomal membrane permeabilization or LMP [43]. Once at the lysosomal membrane, LRRK2 triggers a unique tubulation and sorting event we termed LYTL for LYsosomal Tubulation/sorting driven by LRRK2 [39]. LYTL comprises the deformation of the lysosomal membrane in a tubular structure that elongates through microtubules. These tubules can undergo fission, resulting in sorted vesicles that travel around the cell, interacting with healthy, active lysosomes. Here, we will review our current understanding of LYTL, its regulation and potential cellular function.

LRRK2 recruitment to damaged lysosomes

It has been established that under resting conditions, LRRK2 primarily exhibits a cytosolic and diffuse distribution, without apparent association with any specific organelle or cellular compartment [29]. However, in some cells, LRRK2 can be present at the membrane of a few lysosomes [39,41,44]. The LRRK2-positive lysosomes show decreased degradative capacity, and low levels of the lysosomal hydrolase Cathepsin B. Based on these observations, we reasoned that LRRK2 is recruited to damaged lysosomes. The addition of certain lysosomotropic reagents that induce LMP, including L-leucyl-L-leucine methyl ester (LLOME) [39,41,44–46], Chloroquine (CQ) [40,47] or Nigericin [44,48], increase the recruitment of LRRK2 to lysosomes, supporting our hypothesis. Lysosomotropic reagents are not the only stimuli capable of inducing LRRK2 presence on lysosomes. In RAW264.7 cells, exogenous cellular stressors like bacterial infection [41] or the stimulation of the danger-sensing pathway cyclic GMP–AMP synthase (cGAS)–stimulator of interferon genes (STING) [44] also increase LRRK2 presence on lysosomes. Cells expressing the PD mutation VPS35-D620N [49] also show enhanced LRRK2 recruitment to lysosomes [50]. VPS35 is a member of the retromer complex [51,52] and is responsible for receptor recycling from endosomes to the trans-Golgi network and the plasma membrane. The VPS35-D620N effect on LRRK2 recruitment to lysosomes is likely due to reduced autophagy and lower lysosomal degradative capacity [53]. Collectively, these observations demonstrate that both exogenous and endogenous stimuli that stress lysosomes are sufficient to target LRRK2 to these compartments. Furthermore, that similar phenomena can be accessed by events related to inflammation suggests that LRRK2 activation is a physiological event.

The recruitment of LRRK2 to damaged lysosomes appears to be multifactorial (Figure 1). CASM (conjugation of ATG8 to single membranes) is a non-canonical autophagy pathway wherein ATG8 family members are targeted to single membranes from the endolysosomal system [54]. CASM is independent of many of the classical autophagy core proteins but requires the core ubiquitin-like conjugation systems that support ATG8 lipidation to membranes, including ATG5, ATG7 and ATG16L1 [55]. CASM is activated by the vATPase, a lysosomal pump responsible for lysosomal acidification [56,57]. The engagement of the vATPase's two large domains (V0–V1) is necessary and sufficient to recruit ATG16L1 to the lysosomal membrane under several CASM-activating stimuli [54]. Lysosomotropic reagents, bacterial infection and stimulation of the cGAS–STING pathway all activate CASM and CASM inhibition decreases LRRK2 recruitment and activity on damaged lysosomes [44,47]. In addition to CASM, the small GTPase and LRRK2 substrate RAB12, is also important to recruit and activate LRRK2 to damaged lysosomes [58,59]. It is also possible that membrane properties are important for LRRK2 recruitment, as purified LRRK2 can directly bind to liposomes *in vitro*, with preference for highly curved membranes [60]. Thus, recruitment of LRRK2 to the lysosome likely depends on

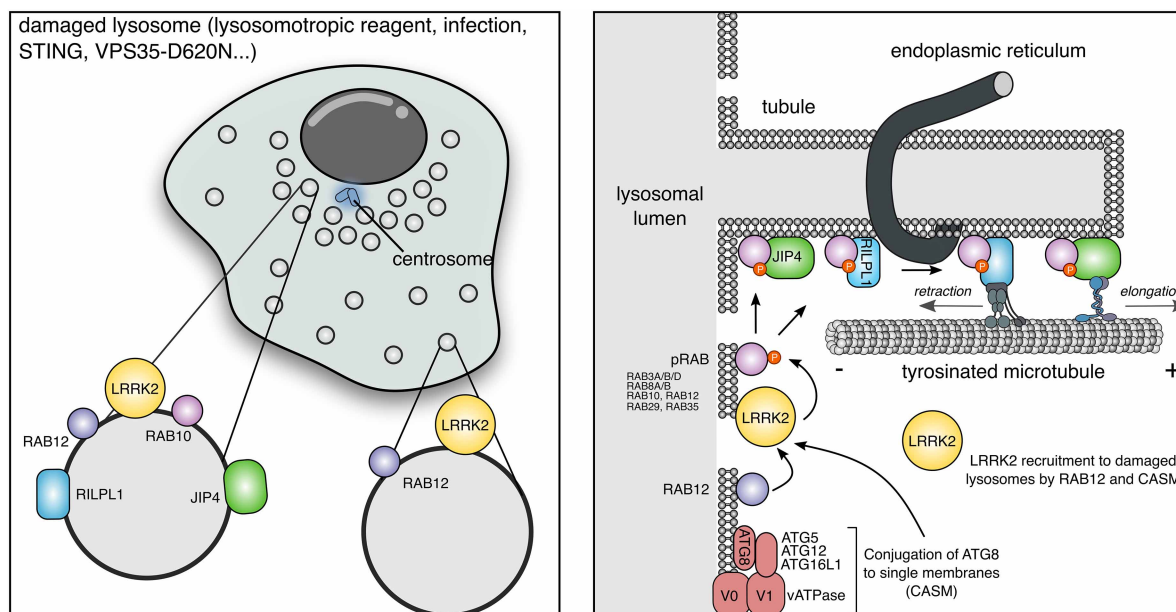


Figure 1. LRRK2 recruitment to damaged lysosomes triggers Lysosomal Tubulation and sorting driven by LRRK2.

(A) Upon different stress stimuli, LRRK2 localizes to the lysosomal membrane where it phosphorylates and recruits a subset of RAB proteins. Cell localization will determine the pattern of recruitment of the pRABs and their effectors. Indeed, pRAB10 and RILPL1/JIP4 will be present almost exclusively on perinuclear lysosomes (clustered around the centrosome). **(B)** LRRK2 is recruited to damaged lysosomes by RAB12 and CASM, and via its pRABs, recruits two RHD proteins (RILPL1 and JIP4). JIP4 binds to an unknown kinesin and elongates tubular structures emanating from lysosomes. RILPL1, by binding the dynactin subunit p150^{Glued}, has the opposite effect retracting the tubular structures toward the minus-end of microtubules. LYTL tubules undergo sorting, partially regulated by the ER, travel around the cell and contact healthy lysosomes.

both membrane and protein components, including some that like vATPase are present basally at lysosomal membranes.

LRRK2 is known to phosphorylate a subset of fourteen RAB GTPases (RAB3A/B/C/D, RAB5A/B/C, RAB8A/B, RAB10, RAB12, RAB29, RAB35 and RAB43) in a conserved region of their switch-II domain [61]. The presence of LRRK2 at a membrane is sufficient to trigger its activation and recruitment of RAB substrates [46,62,63]. Specifically, LRRK2 activation on membranes is independent of the membrane identity [62,63]. Once at the lysosomal membrane, LRRK2 becomes kinase active and recruits and phosphorylates most of its canonical RAB substrates [64]. RABs removal from their membranes require inactivation via hydrolysis of GTP into GDP, and subsequent extraction from the membrane by the GDP dissociation inhibitor (GDI) [65], which solubilizes the inactive prenylated RAB protein into the cytosol. RAB phosphorylation by LRRK2 blocks the ability of RABs to bind to their GDI [61], thus blocking the removal of the RAB from their membrane. Proteomics analysis on isolated lysosomes suggests that LRRK2 recruits nine of its known RAB substrates (RAB3A/B/C, RAB8A/B, RAB10, RAB12, RAB29 and RAB35) (Figure 1) [64], indicating that RAB5A/B/C and RAB43 are not activated by LRRK2 in the context of lysosomal damage.

LRRK2 directs RHD family members to lysosomes via pRAB proteins

RAB GTPases perform their cellular functions by recruiting effectors to the target membrane [66]. Effectors bind a specific RAB, typically in its GTP-bound state, and mediate at least one downstream biological effect [67]. RAB effectors are a diverse group of proteins that affect various cellular functions including membrane tethering, organelle transport or membrane fusion [68]. Structural and biochemical analyses have shown that phospho-RABs (pRABs) can bind RILP-homology domain (RHD) family members [69]. The RHD family members include JNK-interacting protein 3 (JIP3), JNK-interacting protein 4 (JIP4), RILP-like protein 1

(RILPL1) and RILP-like protein 2 (RILPL2). This group of proteins share two RHDs, RHD1 and RHD2 [70] (Figure 2). The phosphorylated substrate on the pRABs binds to an exposed arginine in the N terminal region of the RHD2 [69]. A nearby arginine, which is absent from the related protein RILP, is crucial in stabilizing the interaction [69,71] (Figure 2). The RAB:RHD complex is a heterotetramer, with RHD family members forming a central α -helical dimer that connects two pRAB molecules. The N termini of the α helices form an X-shaped cap that directs arginine residues from RHD proteins toward the phosphorylated residue on the RAB (i.e. threonine 73 for RAB10) (Figure 2).

At the surface of damaged lysosomes, pRABs can recruit at least two members of this family (JIP4 and RILPL1) (Figure 1). JIP4 and RILPL1 stain for tubular structures that emanate from the lysosome and lack typical lysosomal membrane markers and also lack LRRK2 [39,45,64]. These tubules elongate on tyrosinated microtubules [64] and are highly dynamic [45], as they can retract toward the lysosomal membrane or undergo sorting. Based on these observations, we named this process LYTL for LYsosomal Tubulation/sorting driven by LRRK2. Once sorted, LYTL vesicles move around the cytosol and can contact healthy lysosomes [39]. The length of these contacts is variable and has not been shown to result in fusion. We think that these contacts could result in cargo transfer from LYTL vesicles to active lysosomes for degradation, although this hypothesis has not yet been proven empirically.

RHD members typically act as motor adaptor proteins by recruiting motors to different organelles, promoting their movement [70,72]. JIP3 and JIP4 can bind to both dynein-light intermediate chain and kinesin-heavy chain (KIF5) via their RHD1; and to p150^{Glued} and kinesin-light chain via their leucine-zipper II domain (LZII) [70,73–79]. p150^{Glued} is the largest dynactin subunit and is crucial for dynein-dependent retrograde transport [80–82]. RILPL2 is involved in actin-dependent transport, recruiting myosin-Va [83,84], also through its RHD1 [85]. RILPL1 binds to p150^{Glued} and promotes the clustering of LRRK2-positive lysosomes toward the centrosome [64]. LYTL tubules are highly dynamic structures [45] that elongate toward the plus-end of microtubules and retract toward its minus-end [64]. JIP4 depletion reduces tubulation [39], whereas RILPL1 deficiency increases the length of LYTL tubules [64]. These contradictory roles suggest that JIP4 elongates LYTL tubules via an unknown kinesin, and RILPL1 retracts tubules via dynactin/dynein (Figure 1). JIP4 can also bind to and activate dynein when recruited to autophagosomes by ARF6 [86], promoting retrograde autophagosomal transport. In cells expressing hyperactive LRRK2, JIP4 increases its binding to kinesin via RAB10, hampering autophagosomal transport [86,87]. Such observations are consistent with our working model, where pRAB proteins facilitate JIP4 binding to kinesins leading to tubule elongation. We believe that the simultaneous binding of pRABs to RHD proteins and hence to motor proteins that drive transport in opposite directions along microtubules results in dynamic instability of the tubule-associated membrane [64]. Depletion of KIF5B has no effect on LYTL tubulation [64], suggesting that JIP4 binds to a different kinesin during LYTL. The nature of LYTL's kinesin remains unknown, and needs to be addressed in future work. JIP4 binding to dynein can also occur on lysosomes, independently of LRRK2, leading to retrograde lysosomal transport in resting conditions, under starvation or oxidative stress [88–90].

Tubulation and sorting processes in the cell often require the binding of Bin/amphiphysin/Rvs (BAR) domain proteins that promote membrane curvature, allowing tubule formation and subsequent fission [91]. Even though no BAR family member has yet been identified in LYTL [39], LRRK2 GTPase activity can promote tubulation in isolated liposomes [60]. As LRRK2 is absent on LYTL tubules, it is possible that LRRK2 GTPase activity contributes to membrane deformation prior to tubule elongation. More work needs to be done to concretely identify the mechanism responsible for membrane curvature.

The ER consists of a large and interconnected network of tubules and flat cisternae [92]. The ER forms contacts with several cellular structures, which can lead to fission [93–95]. ER tubules establish contacts with LYTL tubules preceding membrane fission and reducing the amount of ER tubules impairs sorting [45]. This observation suggests that the ER promotes sorting due to contacts with LYTL tubules. LYTL is also regulated by lysosomal positioning (Figure 1) [46]. By tagging LRRK2 to a lysosomal targeting sequence we artificially expressed LRRK2 on lysosomes, bypassing the need to add a lysosomotropic reagent or toxic stimuli. Interestingly, pRAB10 and JIP4 are only present in LRRK2-positive perinuclear lysosomes despite LRRK2 being kinase active also in peripheral lysosomes. Manipulation of lysosomal positioning has a similar effect, there is a decrease in pRAB10/JIP4 on LRRK2-positive lysosomes by induction of lysosomal anterograde transport and an increase pRAB10/JIP4 levels on lysosomes by induction of lysosomal retrograde transport. This local regulation of LYTL likely depends on phosphatases or RAB GAPs/GEFs that have differential lysosomal presence depending on lysosomal location, although these proteins remain to be identified.

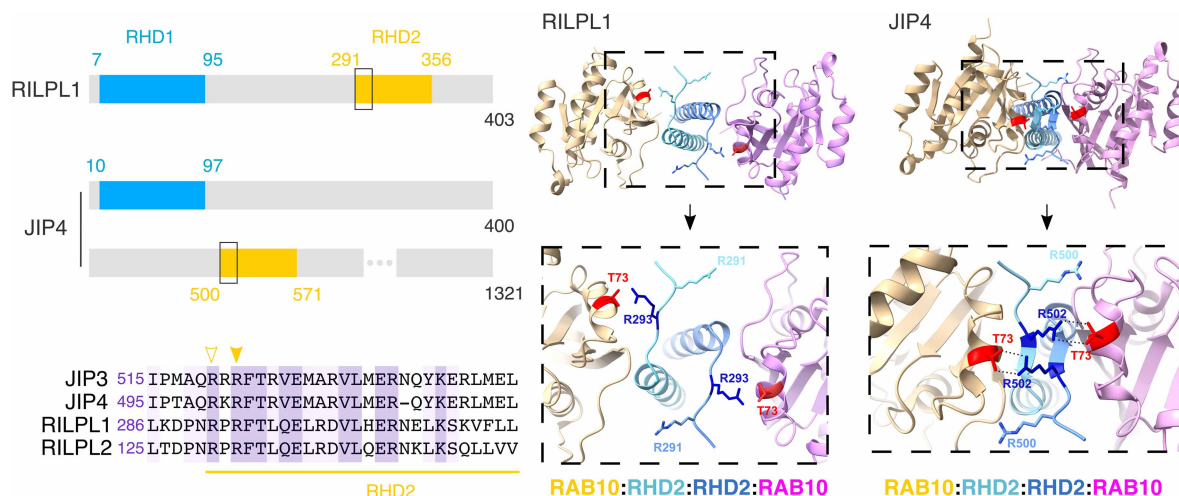


Figure 2. RILP-homology domain (RHD) proteins act as pRAB effectors by binding to the phosphorylated residue through a conserved arginine in their RHD domain.

RILPL1 and JIP4 and both RHD proteins are recruited to the lysosomal membrane via pRAB proteins. Using AlphaFold2 Multimer, we modeled the interaction between a RAB substrate (RAB10) and RILP1/JIP4. The RAB:RHD complex is a heterotetramer, with RHD family members forming a central α -helical dimer that connects two pRAB molecules. The N termini of the α helices form an X-shaped cap that directs arginine residues from RHD proteins toward the phosphorylated residue on the RAB (i.e. threonine 73 for RAB10).

Other lysosomal tubulation events

Lysosomes deform their membranes as a common adaptive response to multiple stimuli [96]. These lysosomal tubulation and membrane remodeling events play a vital role in maintaining cellular homeostasis. We review these different tubules to emphasize that these are morphologically, molecularly and functionally distinct from LYTL.

Tubular lysosomes

Lysosomes can transition from a vesicular to a tubular morphology [97]. This change in morphology is significant during immune cell stimulation [98], as it enhances lysosomal surface area, facilitating MHC-II presentation in dendritic cells [99,100] and promoting phagocytosis in macrophages [101]. Tubular lysosomes (TLs) are also abundant in osteoclasts and muscle cells, regulating bone metabolism [102] and muscle remodeling [103]. Dietary restriction has been shown to promote longevity in various species, such as *Caenorhabditis elegans*, where aging is associated with increased TLs [104]. This morphological change appears to be essential for the longevity induced by starvation in these organisms [105,106].

Lysosomal reformation

To replenish the lysosomal pool during periods of high catabolic activity, including starvation [107], endocytosis [108] or phagocytosis [109], healthy and active lysosomes protrude their membranes into tubular structures that fission into new lysosomes. This highly regulated process is called lysosomal reformation (LR) [110–115]. Notably, LR tubules are stained with LAMP1 and other common lysosomal membrane markers. Although they are different cellular processes, LYTL and LR do share some similarities: (i) the ER also marks both LR and LYTL fission sites [116], underscoring the importance of the ER in promoting membrane fission in multiple cellular structures; and (ii) both elongate toward the plus-end of microtubules, even though depletion of KIF5B, the motor protein for LR [117], has no effect on LYTL tubulation.

ATG8-positive tubules

As discussed above, lysosomal membrane damage induces CASM, a process where ATG8s (LC3A/B/C, GABARAP and GABARAPL1/2) are directed to the membrane of ruptured lysosomes. Once on lysosomes, one

of the ATG8 members (LC3A) physically binds to ATG2A/B [118], which are lipid transfer proteins that have been associated with lysosomal membrane repair by transferring phosphatidylserine to ruptured lysosomes, via the phosphoinositide-initiated membrane tethering and lipid transport (PITT) pathway [119]. Additionally, ATG8s form tubular structures that elongate from lysosomes via microtubules and lack LAMP1 [118]. Similar to LYTL, ATG8-positive tubules are dynamic and can also be sorted into vesicles. ATG8-positive tubules appear unrelated to LYTL, as LRRK2 kinase inhibition has no effect on ATG8 tubulation [118]. The role of these ATG8-positive tubules and vesicles remains unclear and requires future work. However, LYTL and ATG8 tubulation seem to act in parallel during lysosomal membrane damage and could be functionally related.

Conclusions

Lysosomes reform their membranes to respond to membrane damage [12]. Notably, lysosomes stabilize and repair their membranes through multiple different mechanisms that may be partially redundant [119–123]. If damage persists, the cell clears damaged lysosomes via lysophagy [124–132]. LYTL is an alternative process used by lysosomes to respond to more limited membrane damage. Cells that lack LRRK2 or are treated with an LRRK2 kinase inhibitor fail to display recruitment of LYTL proteins (RABs, JIP4 or RILPL1) to lysosomes, suggesting that LRRK2 kinase activity is necessary for LYTL. In culture, the induction of lysosomal damage triggers LRRK2 recruitment to lysosomes, but it is not yet clear if this process can be fully recapitulated in a more physiological context. Recent work has shown that cGAS-STING stimulation can also induce LRRK2 recruitment to lysosomes in a macrophage cell line [44], demonstrating that non-lysosomotropic reagents can also promote LRRK2 presence on lysosomes. Nonetheless, LRRK2 recruitment and activation on lysosomes still needs to be reported in an *in vivo* context.

Once on lysosomes, LRRK2 initiates LYTL via recruitment and phosphorylation of RAB proteins. Our current understanding of LYTL suggests that LRRK2 recruits nine RABs [64] and at least one of them (RAB12) is likely to contribute to LRRK2 recruitment [58,59]. We have shown that RAB10 physically binds to the RHD family members JIP4 and RILPL1, but the role of the remaining RAB substrates in LYTL remains unclear. Interestingly, RAB8A may play a role in iron homeostasis once recruited by LRRK2 to damaged lysosomes [133].

We have shown that the ER contributes to tubule sorting [45], indicating there is likely a dynamin-like protein that remains to be identified playing the role as membrane ‘scissor’. Unpublished data from our laboratory found lack of colocalization of several proteins known to fission membranes in other cellular contexts (DNM2, DNM2, DRP1, EHD1, EndoA2 and VPS4A). MROH1, a recently discovered highly conserved lysosomal fission protein [114], could be a good candidate for LYTL tubule fission.

Once sorted, LYTL vesicles travel around the cell via microtubules and can contact healthy/active lysosomes. We hypothesize that LYTL is a healthy stress response, aimed at targeting undegraded cargo into vesicles for degradation on healthy/active lysosomes. However, this hypothesis remains to be supported or disproven as the cargo transported by LYTL vesicles awaits identification. Another possible hypothesis is that LYTL vesicles could fuse with the plasma membrane, unloading their cargo into the extracellular space. It is also noteworthy that LRRK2 has been proposed to play a role in other organelles or cellular structures (for a review: [29]), further complicating the current knowledge on disease mechanisms in LRRK2-PD. Understanding the cellular role of LYTL is crucial not only to pinpoint the effect of LYTL in cellular homeostasis during lysosomal membrane damage, but specially to better understand the role of LYTL in PD.

Perspectives

- Cells have multiple overlapping ways to respond to lysosomal damage, including Lysosomal Tubulation/sorting driven by LRRK2, showing that avoiding the misregulation of lysosomes and their contents is critical for cell health.
- We have outlined the key mechanistic steps by which LRRK2 affects lysosomal morphology in cells that link known substrates, specifically RAB proteins, to downstream adaptor proteins and molecular motors.

- Further work is required to understand the extent to which we can generalize LYTL across different physiological triggers and identify downstream consequences for the cell.

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Competing Interests

J.H.K. is an employee of Denali Therapeutics Inc.

Abbreviations

BAR, Bin/amphiphysin/Rvs; CASM, conjugation of ATG8 to single membranes; cGAS, cyclic GMP–AMP synthase; ER, endoplasmic reticulum; GDI, GDP dissociation inhibitor; JIP3, JNK-interacting protein 3; JIP4, JNK-interacting protein 4; LR, lysosomal reformation; LRRK2, leucine-rich repeat kinase 2; LYTL, Lysosomal Tubulation/sorting driven by LRRK2; LZII, Leucine-zipper II domain; mTOR, mechanistic target of rapamycin; PD, Parkinson's disease; pRABs, phospho-RABs; RHD, RILP-homology domain; RILPL1, RILP-like protein 1; RILPL2, RILP-like protein 2; STING, stimulator of interferon genes; TLs, tubular lysosomes; vATPase, vacuolar (H⁺) ATPase.

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